

COURSE SYLLABUS

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS (BSIS)

DATA MINING

Year & Section:	4th Year – Section E
Course	DATA MINING
Course Code	ADV08
Program	Bachelor of Science in Information Systems (BSIS)
Year & Section	4th Year – Section E
Course Type	Major / Professional Course
Suggested Term	As prescribed in the approved BSIS curriculum
Credit Units	3 Units
Prerequisite	As prescribed in the approved BSIS curriculum

I. Course Description

This course introduces the concepts, techniques, and practical applications of data mining for discovering useful patterns and knowledge from structured and unstructured data. Students explore data preparation, exploratory data analysis, classification, clustering, association rule mining, anomaly detection, feature selection, model evaluation, and data visualization. The course emphasizes the responsible interpretation and use of mined results in supporting organizational decision-making and Information Systems applications.

II. Course Learning Outcomes

ILO At the end of the course, students should be able to:

ILO 1 Explain fundamental concepts, processes, tasks, and applications of data mining.

ILO 2 Prepare, clean, transform, and explore datasets for data mining activities.

ILO 3 Apply appropriate data mining techniques such as classification, clustering, and association rule mining.

ILO 4 Select and use suitable tools and algorithms to analyze datasets and discover meaningful patterns.

ILO 5 Evaluate data mining results using appropriate performance measures and interpret findings in context.

ILO 6 Develop a basic data mining solution or project that addresses an Information Systems or organizational problem.

ILO 7 Present data mining findings clearly and observe ethical, privacy, security, and responsible data-use principles.

III. Weekly Topics and Syllabus Mapping

Week	Topic / Lesson	Learning Activity	Assessment
1	Introduction to Data Mining	Discuss data mining concepts, KDD process, applications, and course expectations	Diagnostic activity / participation
2	Data Mining Process and Data Understanding	Examine data sources, data types, attributes, and the data mining workflow	Data understanding activity
3	Data Preprocessing	Clean, integrate, transform, and reduce data for analysis	Preprocessing exercise
4	Exploratory Data Analysis and Visualization	Explore datasets and identify patterns using descriptive statistics and visualizations	EDA output / visualization
5	Data Mining Tools and Environments	Explore appropriate software tools, libraries, datasets, and mining environments	Tool-based activity
6	Classification Fundamentals	Explain classification concepts, training/testing data, and decision-making rules	Classification exercise
7	Classification Algorithms	Apply selected classification algorithms and compare basic results	Algorithm activity
8	Midterm Data Mining Evaluation	Demonstrate progress and evaluate preprocessing and initial mining results	Midterm project evaluation
9	Clustering	Apply clustering concepts and techniques to discover natural groupings in data	Clustering output
10	Association Rule Mining	Discover frequent itemsets and association rules using support, confidence, and related measures	Association rule exercise

Week	Topic / Lesson	Learning Activity	Assessment
11	Anomaly and Outlier Detection	Identify unusual observations and discuss possible applications and limitations	Outlier analysis
12	Feature Selection and Model Improvement	Select relevant features and examine their effect on mining performance	Feature selection activity
13	Model Evaluation and Interpretation	Evaluate mining models using suitable metrics and interpret results in context	Evaluation report
14	Data Mining Applications in Information Systems	Examine applications in business intelligence, customer analysis, risk, education, and operations	Application case analysis
15	Ethical, Privacy, and Responsible Data Mining	Discuss privacy, bias, security, transparency, and responsible use of mined information	Ethics / privacy checklist
16	Final Data Mining Project Presentation	Present, demonstrate, and explain the completed data mining project and findings	Final presentation / demonstration

III. Weekly Topics and Syllabus Mapping (Continued)

Week	Topic / Lesson	Learning Activity	Assessment
16	Final Data Mining Project Presentation	Present, demonstrate, and explain the completed data mining project and findings	Final presentation / demonstration

IV. Learning Resources

- Course modules and instructor-provided materials
- Data mining software, programming environments, and analytical tools
- Public and instructor-provided datasets for learning activities
- Documentation for selected data mining libraries, tools, and platforms
- Resources on statistics, machine learning, data visualization, and database systems
- Testing, evaluation, reporting, and project documentation templates
- Institutional and course-specific project guidelines

V. Assessment Methods

Assessment Component	Weight
Laboratory activities and data mining exercises	25%
Data preprocessing, analysis, and mining activities	20%
Data mining project development and implementation	25%
Model evaluation and interpretation	10%
Documentation / technical report	10%
Final presentation and demonstration	10%
TOTAL	100%

VI. General Requirements

- Students are expected to participate actively in lectures, laboratory activities, consultations, and collaborative data analysis tasks.
- Students must complete assigned data mining exercises and maintain appropriate datasets, scripts, outputs, and documentation.
- Students are expected to perform appropriate data preprocessing, testing, evaluation, interpretation, and refinement activities.
- Students must observe academic integrity and properly acknowledge datasets, third-party code, libraries, documentation, and other resources used.
- Use of AI-assisted analytical or development tools must follow instructor and institutional guidelines, and students must be able to explain and demonstrate submitted work.
- Students must observe appropriate privacy and security practices when handling datasets and must not disclose protected or unauthorized information.
- Final project outputs and supporting documents must be submitted according to the instructor's approved schedule.

VII. References

- Course modules and instructor-provided materials.
- Official documentation for selected data mining, statistical, visualization, and machine learning tools.
- Data mining, machine learning, database systems, and Information Systems references identified by the instructor.
- Selected academic articles and reputable materials related to data mining applications.
- Publicly available datasets and documentation appropriate for instructional use.
- Institutional guidelines and other resources identified by the instructor.

ADV08 – DATA MINING

BSIS 4th Year – Section E

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